



CHEMISTRY SSC-I

Inclusive Scheme of Studies 2024 (1200) Marks

SECTION – A (Marks 12)

Time allowed: 20 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Version No.				
1	0	0	5	1

0	●	●	0	0
●	1	1	1	●
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7	7	7	7	7
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9	9	9	9	9

ROLL NUMBER						

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7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Chemistry mainly deals with the study of:	Living organisms	Matter and its interactions	Planets and stars	Historical events	☐	☐	☐	☐
2. Avogadro's constant represents the number of molecules in:	0.1 g of H ₂	14 g of N ₂	44 g of CO ₂	10 g of H ₂ O	☐	☐	☐	☐
3. The sum of oxidation numbers of all atoms in a sulfate ion (SO ₄ ²⁻) must be:	-2	-1	0	+2	☐	☐	☐	☐
4. Dissolving ammonium nitrate in water causes the container to feel cold. This shows the process is:	Neutralization	Exothermic	Endothermic	Combustion	☐	☐	☐	☐
5. For a reversible reaction, why might the reaction not go to completion?	Products never form	Forward reaction is faster than backward	Backward reaction occurs simultaneously	Energy is not involved	☐	☐	☐	☐
6. Which reaction products correctly show what happens when an acid reacts with a metal carbonate?	Salt + CO ₂ + H ₂ O	Salt + H ₂ + NH ₃	Salt + N ₂ + H ₂ O	Salt only	☐	☐	☐	☐
7. An element in Group I-A forms ions with charge:	-1	+1	+2	-2	☐	☐	☐	☐
8. Which trend is observed down the Group in Group I-A elements?	All become gases	Melting point increases	Reactivity decreases	Density increases	☐	☐	☐	☐
9. The approximate percentage of nitrogen (N ₂) in clean, dry air is:	10%	21%	78%	90%	☐	☐	☐	☐
10. Sulfur dioxide (SO ₂) is released into the air during:	Use of catalytic converters in cars and vehicle	Combustion of fossil fuels containing sulfur compounds	Melting of ice on glaciers	Rusting of iron in industry	☐	☐	☐	☐
11. Which of the following can have more than one structural formula?	CH ₄	C ₂ H ₆	C ₃ H ₈	C ₄ H ₈	☐	☐	☐	☐
12. An alkane is most likely to undergo:	Addition reaction	Neutralization reaction	Dehydration reaction	Substitution reaction	☐	☐	☐	☐

SUPPLEMENTARY TABLE

—1SA-I 26005-(B)—

Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20



CHEMISTRY SSC-I

Inclusive Scheme of Studies 2024 (1200) Marks

Time allowed: 2:40 Hours

Total Marks Sections B and C: 48

SECTION – B (Marks 30)

Q. 2 Attempt the following parts.

(10 x 3 =30)

(i)	Identify the branch of Chemistry involved in the given statements: a. Large scale production of NH_3 by Haber's process. b. Breakdown of glucose gives energy to our body. c. Use of chloroflouro carbons (CFCs) in refrigerator linked to depletion of ozone layer.	1+1 +1	OR	Show the following conversions by drawing their electrons dot structures: a. $\text{Na} \rightarrow \text{Na}^+$ b. $\text{Cl} \rightarrow \text{Cl}^-$	1.5 + 1.5
(ii)	Why solids have high density and very low compressibility whereas gases have opposite behavior? Explain briefly.	1.5 + 1.5	OR	Calculate the number of moles in a foil of Aluminum that weights 100 g.	03
(iii)	An element exists as two isotopes: • ^{64}X (abundance 30%) • ^{66}X (abundance 70%) Calculate the average relative atomic mass of element X	03	OR	Determine the number of Protons and Neutrons in given Isotopes: a. $^{23}_{11}\text{Na}$ b. $^{37}_{17}\text{Cl}$ c. $^{13}_6\text{C}$	1+1 +1
(iv)	Identify the given substances as solution, suspension or colloidal: a. Air b. Sand in water c. Jelly	1+1 +1	OR	Briefly describe: a. How a catalyst affects the reaction rate? b. How a catalyst affects the activation energy?	1+2
(v)	What is meant by 'element'? State one distinguishing feature that separates it from mixtures.	2+1	OR	How is ionic bond formed? Give one example of a compound having ionic bond.	2+1
(vi)	Using the bond energies given below, calculate the enthalpy change (ΔH) for the reaction: $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ Bond energies: $\text{H}-\text{H} = 436 \text{ kJ/mol}$, $\text{Cl}-\text{Cl} = 242 \text{ kJ/mol}$ and $\text{H}-\text{Cl} = 431 \text{ kJ/mol}$	03	OR	Justify the following statements: a. Flourine has smaller shielding effect than Chlorine b. Flourine has high ionization energy than Chlorine	1.5 + 1.5
(vii)	How is solubility affected by increasing temperature for given solutions: a. KCl in water (endothermic solution) b. $\text{Ca}(\text{OH})_2$ in water (exothermic solution)	1.5 + 1.5	OR	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}_{(\text{blue solid})} \rightleftharpoons \text{CuSO}_{4(\text{white solid})} + 5\text{H}_2\text{O}$ How heating will affect the equilibrium in given reaction?	03
(viii)	Briefly state the trend of metallic character along the group. Also give reason.	1.5 + 1.5	OR	Write chemical reaction when an acid reacts with carbonate. Describe the visible change that may appear.	2+1
(ix)	State the general trends observed down the group for elements of Group-I for their density, and reactivity.	1.5 + 1.5	OR	Name the three main types of chemical fertilizers commonly used in agriculture.	1+1 +1
(x)	How are 'dissolved oxygen' and 'essential minerals' beneficial for aquatic life? Describe briefly.	1.5 + 1.5	OR	What are substitution reactions? Which type of compounds shows these reactions?	2+1

SECTION – C (Marks 18)

Note: Attempt the following questions.

(3x6=18)

Q.3	Show covalent bond formation in given molecules using dot-and-cross structures: a. CO_2 b. H_2O c. CH_4	2+2 +2	OR	What is meant by 'molecular formula' and 'structural formula'? Give one example each. Show that $\text{C}_3\text{H}_6\text{O}_2$ can have more than one structural formula.	2+2 +1
Q.4	Explain the terms 'Acid' and 'Base' according to the Bronsted-Lowry theory. Identify the acid and base in the reaction: $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$	2+2 +2	OR	Discuss the functions of carbohydrates and proteins in human body. (any three specific functions each)	3+3
Q.5	Compare metals and non-metals in terms of their: a. Thermal conductivity b. Electrical conductivity c. Malleability and ductility	2+2 +2	OR	How ethane can be prepared by: a. Hydrogenation of alkynes b. Reduction of alkyl Halides	3+3

SUPPLEMENTARY TABLE

— 1SA-I 26005 (B) —

Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20



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9	9	9	9	9	9	9

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Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	سوال	A	B	C	D	A	B	C	D
1. Which state of matter is represented by air?		Solid	Liquid	Gas	Plasma	☐	☐	☐	☐
2. Diamond and graphite are allotropic forms of:		Oxygen	Carbon	Silicon	Sulfur	☐	☐	☐	☐
3. In the given reaction $2Na + Cl_2 \rightarrow 2NaCl$ sodium changes its oxidation state from 0 to +1, it is the process of:		Oxidation	Reduction	Elimination	Addition	☐	☐	☐	☐
4. A reaction that does not go to completion is called:		Irreversible	Endothermic	Reversible	Exothermic	☐	☐	☐	☐
5. When a base reacts with an ammonium salt, it produces:		Hydrogen	Carbon dioxide	Oxygen	Ammonia	☐	☐	☐	☐
6. The charge on an ion depends on the number of:		Neutrons	Protons	Valence electrons	Atomic mass	☐	☐	☐	☐
7. Which of the following relationships is INCORRECT regarding relative reactivity of Halogens for displacement reactions?		$F_2 > Cl_2$	$Br_2 > F_2$	$Br_2 > I_2$	$Cl_2 > Br_2$	☐	☐	☐	☐
8. The percentage of oxygen in clean, dry air is about:		21%	50%	78%	92%	☐	☐	☐	☐
9. Which factor increases water scarcity in Pakistan?		Population growth	Cloud formation	Wind speed	Soil erosion	☐	☐	☐	☐
10. The general formula of Carboxylic acids is:		$C_nH_{2n+2}COOH$	$C_nH_{2n}O_2$	$C_nH_{2n+1}COOH$	$C_nH_{2n}COOH$	☐	☐	☐	☐
11. Fish is a good source of:		Carbohydrate	Sugar	Protein	Glucose	☐	☐	☐	☐
12. Reduction of propyl chloride mainly produces:		Propene	Propane	Methane	Propyne	☐	☐	☐	☐

—1SA-I 26005 (D)—

SUPPLEMENTARY TABLE

Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20



CHEMISTRY SSC-I

Inclusive Scheme of Studies 2024 (1200) Marks

Time allowed: 2:40 Hours

Total Marks Sections B and C: 48

SECTION – B (Marks 30)

Q. 2 Attempt the following parts.

(10 x 3 =30)

(i)	Identify the branch of chemistry that deals with given statements: a. According to Boyle's law when pressure on a gas is applied its volume reduces. b. Industries produce SO ₂ which cause acid rain. c. Our hair has a protein called Keratin which is essential for hair growth.	1+1 +1	OR	Differentiate between suspension and colloidal with one suitable example each.	1.5 + 1.5
(ii)	How do the 'energy' and 'distance' from the nucleus change as the shell number increases?	1.5 + 1.5	OR	Which type of reaction has a positive sign of enthalpy change? Give one example for that reaction.	1+2
(iii)	Which sub-atomic particle determines the atomic number of an element? What is the significance of atomic number?	1+2	OR	State the trend in melting point and density of Group I-A elements down the group.	1.5 + 1.5
(iv)	Justify the following statements: a. An atom is electrically neutral. b. An atom can convert into positive ion when it loses electron.	1.5 + 1.5	OR	What is meant by activation energy? How does a catalyst affect the activation energy of a reaction?	2+1
(v)	Find the mass of 0.5 moles of NaCl.	03	OR	Derive the chemical formula of the compound formed between: a. Mg ²⁺ and Cl ⁻ c. Ca ²⁺ and NO ₃ ⁻¹ b. Al ³⁺ and SO ₄ ⁻²	1x3
(vi)	$CoCl_2 \cdot 6H_2O_{(pink\ solid)} \rightleftharpoons CoCl_2_{(blue\ solid)} + 6H_2O$ Describe the effect of addition of water on equilibrium in given reaction.	03	OR	Write the dissociation equations of sulfuric acid (H ₂ SO ₄) and sodium hydroxide (NaOH) in aqueous solution.	1.5 + 1.5
(vii)	An element has electronic configuration 2,8,7 (atomic number=17). Identify its period, group number and number of valence electrons.	1+1 +1	OR	Complete the given displacement reactions of Halogens if they are possible: a. Br ₂ + 2KI → c. I ₂ + 2KBr → b. F ₂ + 2KCl →	1x3
(viii)	Write benefits of dissolved oxygen and essential minerals found in water.	1.5 + 1.5	OR	Classify the following compounds as alcohol, ether and phenol: a. CH ₃ OCH ₂ CH ₃ c. C ₆ H ₅ OH b. CH ₃ OH	1+1 +1
(ix)	Why are Sedimentation and Chlorination used in water treatment?	1.5 + 1.5	OR	What is meant by 'homologous series' in organic chemistry? Why do members of a homologous series have similar chemical properties?	2+1
(x)	Why are alkanes called saturated hydrocarbons? Write their general formula to determine their molecular formula.	2+1	OR	What is meant by nutrition? Why is healthy eating important for the human body?	1+2

SECTION – C (Marks 18)

Note: Attempt the following questions.

(3x6=18)

Q.3	Differentiate between compounds, and mixtures with respect to: a. Composition b. Chemical bonding c. Separation methods	2+2 +2	OR	Explain the importance of lipids as biomolecules, by mentioning their specific sources and functions in the body. (any three each)	06
Q.4	Show the ionic bond formation of following by using electron dot-and-cross structures: a. NaF b. CaCl ₂	3+3	OR	An element Z is located below Magnesium in Group II-A: a. Predict its relative reactivity in comparison to Magnesium b. Compare its metallic character with Magnesium c. Compare its Ionization energy with Magnesium	2x3
Q.5	Write chemical reactions of HCl when it reacts with: a. Mg b. CaCO ₃ c. NaOH	2x3	OR	What are substitution reactions? Why alkanes show substitution reactions? Illustrate the monosubstituted product formed from Methane when it reacts with Cl ₂ .	2+2 +2

SUPPLEMENTARY TABLE

— 1SA-I 26005 (D) —

Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20



CHEMISTRY SSC-I

Inclusive Scheme of Studies 2024 (1200) Marks

SECTION – A (Marks 12)

Time allowed: 20 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Version No.				
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9	9	9	9	9	9	9

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Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Which part of the atom contains protons and neutrons?	Electron cloud	Shells	Nucleus	Orbits	☐	☐	☐	☐
2. Which type of compounds generally have low boiling points?	Ionic compounds	Network solids	Metallic compounds	Covalent compounds	☐	☐	☐	☐
3. Which compound is formed by Al^{3+} and O^{2-} ?	AlO	Al_2O	Al_2O_3	Al_3O_2	☐	☐	☐	☐
4. The net ionic equation for forming silver chloride is:	$Na^+ + Cl^- \rightarrow NaCl$	$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$	$Ag^+_{(aq)} + Cl^-_{(aq)} \rightarrow AgCl_{(s)}$	$Ag_{(s)} + Cl_{(g)} \rightarrow AgCl_{(s)}$	☐	☐	☐	☐
5. Which symbol shows a reversible reaction?	$\rightarrow$	$\rightleftharpoons$	$=$	$\leftrightarrow$	☐	☐	☐	☐
6. Which reaction best shows a property of a base?	Base + metal $\rightarrow$ salt	Base + water $\rightarrow$ acid	Base $\rightarrow$ ions only	Base + acid $\rightarrow$ salt + water	☐	☐	☐	☐
7. The period number of an element depends on the number of:	Occupied shells	Electrons	Protons	Neutrons	☐	☐	☐	☐
8. Carbon monoxide and particulates are mainly produced by:	Complete combustion	Incomplete combustion	Plant respiration	Evaporation	☐	☐	☐	☐
9. Sulfur contributes to acid rain mainly by forming:	HNO_3	H_2CO_3	HCl	H_2SO_4	☐	☐	☐	☐
10. Which substance in natural water is useful for aquatic life?	Dissolved oxygen	Sewage	Dissolved Nitrates	Dissolved Plastics	☐	☐	☐	☐
11. Which compound is an alkane?	C_2H_4	C_4H_{10}	C_2H_2	C_3H_6	☐	☐	☐	☐
12. Substitution reactions commonly occur in:	Alkanes	Alkynes	Alkenes	Aldehydes	☐	☐	☐	☐

SUPPLEMENTARY TABLE

—1SA-I 26005-(M)—

Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40
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CHEMISTRY SSC-I

Inclusive Scheme of Studies 2024 (1200) Marks

Time allowed: 2:40 Hours

Total Marks Sections B and C: 48

SECTION – B (Marks 30)

(10 x 3 = 30)

Q. 2 Attempt the following parts.

(i)	Identify the branch of chemistry that deals with statements: a. Temperature of the earth is increasing gradually due to increase in concentration of CO ₂ . b. Contact process is used to manufacture sulphuric acid on large scale. c. When a gas is cooled to –273°C it liquefies.	1+1 +1	OR	How is solubility affected by increasing temperature for given solutions: a. KCl in water (endothermic solution) b. Ca(OH) ₂ in water (exothermic solution)	1.5 + 1.5
(ii)	How do the paths of positive and negative charges differ in the same uniform electric field? Make a sketch.	1+1 +1	OR	What is meant by reversible chemical reaction? Does it ever go to completion? Give reason.	1+1 +1
(iii)	How is carbon-14 used to determine the age of ancient objects? Explain briefly.	03	OR	Calculate the number of molecules present in 2 moles of CO ₂ .	03
(iv)	What is activation energy? What is meant by a successful collision in a chemical reaction?	2+1	OR	State Bronsted-Lowry acid base concept. Give example of reaction between HCl and NH ₃ .	1+1 +1
(v)	$\text{CoCl}_2 \cdot 6\text{H}_2\text{O}(\text{pink solid}) \rightleftharpoons \text{CoCl}_2(\text{blue solid}) + 6\text{H}_2\text{O}$ How is above reaction affected by addition of water?	03	OR	Complete the given displacement reactions if possible: a. $\text{Cl}_2 + 2\text{KF} \rightarrow$ b. $\text{F}_2 + 2\text{KI} \rightarrow$ c. $\text{Cl}_2 + 2\text{KBr} \rightarrow$	1x3
(vi)	What is periodic table? On what basis are elements arranged in the modern periodic table?	2+1	OR	Name three nitrogen-containing fertilizers used in agriculture.	1+1 +1
(vii)	What is meant by the thermal stability of hydrogen halides? Arrange HF, HCl, HBr, and HI in order of increasing thermal stability.	2+1	OR	What is meant by healthy eating? How does it support body functions?	2+1
(viii)	Draw the structural formula of ethene (C ₂ H ₄) and ethanol (C ₂ H ₅ OH) showing the correct bonds.	1.5 + 1.5	OR	Name the three main biomolecules and state one food source of each.	1+1 +1
(ix)	Write any three characteristics of homologous series.	03	OR	Give two examples of carbohydrate-rich foods. Briefly explain the role of carbohydrates in human body.	1+2
(x)	Write chemical equations for the combustion of methane: a. In limited supply of oxygen b. In excess supply of oxygen	1.5 + 1.5	OR	Briefly describe the compressibility of solids, liquids and gases with reason.	1+1 +1

SECTION – C (Marks 18)

Note: Attempt the following questions.

(3x6=18)

Q.3	Compare diamond and graphite in terms of their structure and properties.	3+3	OR	Describe metallic bond in terms of electrostatic attraction. Write any two properties of metals which makes them different from the non-metals.	4+2
Q.4	Identify the oxidizing and reducing agents in given reactions by assigning their oxidation states (H=+1, O= –2) a. $\text{WO}_3 + 3\text{H}_2 \rightarrow \text{W} + 3\text{H}_2\text{O}$ b. $\text{H}_2\text{S} + \text{Cl}_2 \rightarrow 2\text{HCl} + \text{S}$	3+3	OR	Differentiate between strong acid and a weak acid. Write dissociation equations for HCl and CH ₃ COOH in aqueous solution to support your answer.	1+1 +2 +2
Q.5	a. What is meant by exothermic reaction and endothermic reactions? b. Calculate the enthalpy change (ΔH) for given reaction. $\text{H}_{2(g)} + \text{F}_{2(g)} \rightarrow 2\text{HF}_{(g)}$ Bond energy values for H-H, F-F and H-F bonds are 436 kJ/mol, 155 kJ/mol and 567 kJ/mol.	(1.5 + 1.5) +3	OR	Explain in detail how the following properties change down a group in the periodic table for Group-IA elements: a. Atomic radius b. Reactivity Give reasons for each trend.	3+3

SUPPLEMENTARY TABLE

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